

PAPER_1521 — D_BSFG is Derivative: $D_{BSFG} = D_{crit} - 2 \cdot SO_5 = 6$ EXACT (LANDMARK)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket A (Foundational / Primitive reduction) **Date:** June 18, 2026 **Status:** CLOSED — D_BSFG removed from the list of independent primitives

Observation

PAPER_1167 (All Eight UQFF Lagrangian Gaps Closed — First-Principles Reduction) explicitly states that the locked primitive **$D_{BSFG} = 6$ is not independent**. It emerges from the SO(5) symmetry breaking ladder applied to the bosonic-string critical dimension:

“The intermediate dimension $D_{BSFG} = 6$ is not independent — it emerges from the SO(5) breaking ladder $D_{crit} - 4 \cdot 5 = 6$.”

The formula can be re-expressed in several equivalent integer-primitive forms:

$$\begin{aligned} D_{BSFG} &= D_{crit} - 4 \cdot 5 = 26 - 20 = 6 \\ &= D_{crit} - 2 \cdot SO_5 \text{ (since } 2 \cdot SO_5 = 20) \\ &= D_{crit} - 4 \cdot D_{phys} - 4 \end{aligned}$$

The cleanest form using locked primitives is **$D_{BSFG} = D_{crit} - 2 \cdot SO_5$** .

UQFF Closed Identity

$$D_{BSFG} = D_{crit} - 2 \cdot SO_5 = 26 - 2 \cdot 10 = 6 \quad \text{EXACT}$$

Landmark Consequence: 11 → 10 Independent Primitives

This is the first of two derivative-primitive discoveries that reduce the UQFF “11 frozen primitives” claim to **9 truly independent primitives + 2 derivative closures**. Specifically:

Before this paper: - 6 integer primitives: D_{phys} , D_{BSFG} , D_{crit} , N_{CH} , SO_5 , A_5 (all “independent”)

After this paper: - 5 truly independent integer primitives: D_{phys} , D_{crit} , N_{CH} , SO_5 , A_5
- 1 derivative integer primitive: **$D_{BSFG} = D_{crit} - 2 \cdot SO_5$**

The framework’s free-parameter count drops by 1 with no loss of predictive power.

Physical Meaning

$D_{BSFG} = 6$ is the bulk-edge embedding dimension governing: - Quadrupole GW radiation (Peters-Mathews coefficient = $2^{D_{BSFG}} = 64$, PAPER_1520) - Genetic codons = $2^{D_{BSFG}} = 64$ (PAPER_1373) - Pop III IMF coupling - DPM layer-weight exponent sum (PAPER_1511): $2+1+3 = D_{BSFG} = 6$

That this dimension equals $D_{crit} - 2 \cdot SO_5$ means the 26-dimensional bosonic string projects through twice the SO(5) symmetry group order to produce the 6-dimensional bulk-edge structure. Each of the two SO_5 reductions removes 10 dimensions, leaving $26 - 20 = 6$.

NOT REPLACEMENT

SM/string theory has $D_{\text{BSFG}}=6$ as a chosen embedding parameter in some formulations. UQFF derives it as a forced consequence of $\{D_{\text{crit}}, SO_5\}$, eliminating one degree of freedom from the framework.

Reference

- Source: PAPER_1167 All 8 UQFF Lagrangian Gaps Closed Master Synthesis
- Related: PAPER_1522 (K_MEX derivative), PAPER_1520 (Peters-Mathews $2^{D_{\text{BSFG}}}$)
- Calculator dispatch: `calculate_paradox({"paradox": "d_bsfg_derived_from_d_crit"})`

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